Event-Aware Representation Learning for Neutral Multi-Source News Summarization

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Abstract

Recent work on neutral news generation suggests that shaping latent representations may help mitigate framing bias. We propose to investigate whether event identity can serve as the organizing principle of the representation space. Our primary approach will use event-aware triplet learning to bring representations of reports covering the same event closer while separating carefully chosen contrasting reports, jointly with a neutral text generation objective. Cross-report supervision will be used only during training, while inference will operate on a single article. As an exploratory extension, we also plan to study JEPA-style cross-report prediction, where the representation of one same-event report is used to predict another. We will evaluate whether these objectives improve neutrality, event retention, and robustness to framing variation.

1 Project Idea

We propose a neutral multi-source news summarization system whose internal representation is explicitly organised around the *event* being reported rather than the *publisher* reporting it. Given several articles about the same story from outlets of differing political leaning, the model produces a single neutral summary; alongside the usual generation objective, its encoder is trained with a triplet loss that pulls together differently-framed reports of the same event and pushes apart same-publisher reports of different events.

The deliverable is a summarization model with an event-aware representation-learning objective, evaluated on how much event content it retains against how much publisher-specific information remains recoverable from its representations.

Before writing this proposal we profiled every corpus we intend to use and reimplemented the evaluation metric the relevant literature reports. Three of the findings from that audit changed the design, and we state them here because they are the substance of our motivation rather than background detail.

2 Motivation

News reports are not neutral observations of events. Different publishers may emphasize different facts, use different language, omit details, or frame the same occurrence through different editorial or ideological perspectives. This makes neutralization difficult: a model must preserve the underlying event while reducing reliance on source-specific presentation. Existing work suggests that this problem is not purely about output wording. NeutraSum directly shapes latent representations toward neutrality, while event-relation graph methods show that explicitly modeling event information can improve framing-bias mitigation and content preservation. These findings motivate us to explore whether the shared information across differently framed reports can provide a useful learning signal. Rather than treating any single source as an authoritative account, multiple reports of the same occurrence can be viewed as complementary views whose common content may better reflect the information that should be preserved during neutralization.

3 Preliminary Analysis

We have already built the data pipeline and run the CPU-side measurements; the numbers quoted above come from it. Table 1 summarises what each corpus can support, and Table 2 gives our preliminary probe results.

These are the lexical floor. The same probes on frozen encoder representations, before and after our objective, are the core measurement of the

Corpus	Event	Pub.	Neutral
NeuS (3,095 st.)	yes	yes	yes
AllSides (37,554)	no	yes	no
WCEP-10 (10,200)	yes	no	partial
WNC (181k)	no	no	partial

Table 1: Which labels each corpus carries: event grouping, publisher (**Pub.**) and a neutral target. Sizes are stories, articles, clusters and sentence pairs respectively. An event-aware triplet objective needs all three labels on one example; only NeuS supplies them together, and it is the smallest of the four.

TF-IDF probe	Acc.	Base	Cls.
Publisher, random split	17.8	9.2	229
Stance, random split	40.2	37.3	3
Stance, outlet-disjoint	35.4	46.0	3

Table 2: Lexical probes on 9,282 NeuS articles; **Base** is the test-majority baseline throughout and **Cls.** the number of classes. Publisher is recoverable from word counts at roughly twice its baseline. Stance is barely recoverable even with outlets shared, and across held-out outlets it drops below the baseline entirely, so what a lexical model learns here is outlet identity, not ideology.

project: the question is whether the triplet term lowers publisher recoverability from the representation *without* lowering ROUGE.

4 Background and Related Work

NeuS (Lee et al., 2022) introduces the task, the AllSides-derived corpus of left/centre/right article triples with human-written neutral "roundup" summaries, and the framing-bias metric. It fine-tunes BART (Lewis et al., 2020) on the concatenated sources and reports that a multi-task auxiliary objective over the encoder helps, but the auxiliary task is classification, and nothing ties the representation to event identity.

We Can Detect Your Bias (Baly et al., 2020) predicts article-level political ideology and uses a triplet objective to strip publisher identity from a representation while preserving ideology. It also contributes the media-based split protocol, in which whole outlets are held out. We invert the objective: preserve *event* identity while removing *source* information.

Multi-document event relation graphs (Lei and Huang, 2025) target the same framing-bias mitigation goal on the same corpus by building an explicit event relation graph and reasoning over it with an LLM. Our approach is the representation-level counterpart: rather than externalising event structure into a graph, we constrain the encoder's latent space to be organised by event. We regard this as the closest prior work and will compare against it directly.

I-JEPA (Assran et al., 2023) predicts one view's latent representation from another rather than reconstructing raw content; this motivates our label-efficient pretraining extension, where the "views" are two outlets' reports of the same event.

DiverseSumm (Huang et al., 2024) proposes QA-based evaluation of whether a multi-document summary retains information from its sources.

5 Contributions

1. **An event-aware triplet objective** for multi-source neutral summarization, in which the positive shares an event with the anchor and the negative shares a *publisher*. The shared publisher is the mechanism: it is the only configuration in which satisfying the margin requires the encoder to stop relying on outlet style. This addresses the first shortcoming: that standard training leaves the encoder's organising principle unconstrained.

2. **A length-normalised framing-bias metric**, together with the evidence that the standard metric is a length proxy. We will report both, and show what conclusions change. This addresses the third shortcoming, and makes neutrality-by-deletion visible rather than rewarded.

3. **A publisher-disjoint evaluation protocol** for this corpus: a stance-stratified split in which no outlet appears on two sides, plus a leakage probe measured on both the standard and the disjoint split. This addresses the fourth shortcoming: that the released split cannot ask the generalisation question.

4. **A reproducible data pipeline.** The released NeuS split files have their publisher and stance labels stripped. We recover them at 100% coverage by normalising headlines against the union of all three crawl files; the obvious single-file join silently misses 13% of examples. We will release this alongside our code.

6 Proposed Methodology

Backbone. A pretrained sequence-to-sequence model (BART-large). The k articles for an event are concatenated into a single source sequence and decoded into a neutral summary under the usual token-level cross-entropy loss against the human-written roundup.

Event-aware triplet objective. For an anchor article a, we sample a positive p (a different outlet's report of the same event) and a negative n (the same outlet's report of a different event). Each is passed through the shared encoder and pooled at the EOS position to give $f(\cdot)$. The training signal is

$$\mathcal{L} = \mathcal{L}_{\text{gen}} + \lambda \cdot \max\big(0,\ \|f(a) - f(p)\| - \|f(a) - f(n)\| + m\big),$$

with margin m and weight λ swept on the validation set. Because a and n always share a publisher, the only way to satisfy the margin consistently is to stop encoding publisher style and organise around the event instead.

Structurally this attaches at the same point as NeuS's own auxiliary-task hook, which adds a weighted auxiliary loss over a pooled encoder state; we replace their binary classification head with the triplet term and reimplement on a current `transformers` stack rather than their pinned 2021 fork.

Sampling and training details. Triples are mined offline: for each of the 9,282 articles we draw two positives and two negatives, discarding anchors whose outlet appears in only one story, which yields 18,368 triples. The triplet term is fed as a second stream alongside the generation batch, mirroring the reference implementation's separate auxiliary-input path, so the two losses need not share a batch size. We start from the hyperparameters NeuS reports (learning rate 3×10^{-5}, effective batch 16, source length 512, target length 250) and sweep only λ and m. Training is short: 2,452 examples over five epochs is roughly 3,000 optimiser steps, under an hour of single-GPU time per configuration, which is what makes the ablation grid and the multi-seed reporting affordable.

Extension: cross-source pretraining. Following the JEPA principle, we pretrain by predicting one outlet's representation of an event from another outlet's representation of the same event, using unlabelled same-event pairs from WCEP-10, which needs neither publisher labels nor neutral targets, and so suits this objective exactly. The question is whether the decoder can then be fine-tuned on a fraction of the labelled summaries and still match a fully supervised model. This extension is contingent on compute and will be dropped before the core result if necessary.

7 Dataset

Our primary corpus is the NeuS release (Lee et al., 2022): 3,095 AllSides stories, each pairing three articles (one left, one centre, one right) with a human-written neutral roundup, spanning 2016–2021 and 250 distinct outlets. It is the only corpus available to us that carries an event grouping, a publisher label and a neutral target on the same example. From it we derive 9,282 publisher-labelled articles and 18,368 training triples; 153 of the 250 outlets appear in at least two distinct stories, which makes 9,188 of 9,285 article slots usable as a same-publisher negative.

Two properties we flag honestly. First, AllSides publishes an *excerpt* per outlet rather than the full article (a hard cap of 95 words, median 85), so a three-source input is ≈241 words and fits comfortably within BART's 1,024-token window, but there is correspondingly little event content to retain. Second, at 2,452 training examples the corpus is small for fine-tuning BART-large; we will report mean and spread over at least three seeds rather than single runs.

We use two further corpora in supporting roles. WCEP-10 (Gholipour Ghalandari et al., 2020) provides 10,200 genuine multi-document event clusters of ≈9 articles each, far better event structure and scale than NeuS, but its preprocessing stripped the publisher labels and its 25-word targets were never written to be neutral, so we use it only for the pretraining extension. AllSides / Article-Bias-Prediction (Baly et al., 2020) supplies 37,554 publisher-labelled articles for additional negatives and for its prebuilt media-based split. We note that its bias labels are outlet-level: only 7 of its 491 sources ever carry more than one label, so ideology prediction and publisher prediction are near-identical tasks there, and we will not report them as independent axes.

8 Evaluation Plan

Event retention: question–answer coverage. We generate question–answer pairs from the source

articles and measure how many remain answerable from the generated summary, following the QA-based faithfulness protocol of FEQA (Durmus et al., 2020). There are two ways to run this, and they trade off against each other. (i) On our own NeuS-derived evaluation set: this keeps the stance labels, and so is the only option that can ask whether coverage is *balanced* across left, centre and right sources; that is, whether the model systematically drops facts that appear only in the right-leaning article. It is the more expensive option in time and compute. (ii) On DiverseSumm (Huang et al., 2024): purpose-built for this measurement, but its articles carry no ideology labels, so it cannot detect that failure mode, and running different evaluation axes on different corpora would make the comparison unfair. We therefore plan on (i) and treat DiverseSumm as a secondary sanity check.

Tonality. We score generations with the NRC-VAD (Mohammad, 2018) valence–arousal–dominance lexicon, reporting both the framing-bias score as NeuS defines it and the length-normalised affect rate introduced above, so that a score lowered by generating less is distinguishable from one lowered by generating more neutrally.

Publisher leakage. We train a publisher classifier on frozen representations and on output text, and run it over both the source articles and the generated summaries. We expect macro-F1 to fall, ideally to the random baseline. This is measured on both the standard split and our publisher-disjoint split.

Summary quality and ablations. We train two models (a control BART fine-tuned with the generation loss alone, and the same model with the triplet term added) and compare every metric above plus ROUGE across the two, so that any significant effect can be attributed to the objective rather than to the backbone or the data. The λ and margin sweeps give the intermediate points between them.

9 Feasibility and Risks

The compute is modest: the full experimental grid (baseline, triplet sweep, ablations, and reruns on the disjoint split) is roughly 20–25 single-GPU hours at 16 GB, which our cluster allocation covers comfortably. The data pipeline, the metric, and the probes are CPU-only and already run.

Three risks we are managing explicitly. **Small data:** 2,452 training examples will produce noticeable seed variance, so every reported number will be a mean over at least three seeds with spread. **Weak event content:** the 85-word source excerpts limit what an event-retention evaluation can claim, so we have demoted the QA check to a secondary measure rather than build the argument on it. **A null result is possible:** the triplet term may reduce publisher recoverability without improving summary neutrality, or may trade ROUGE for it. We regard that as a reportable outcome; combined with the metric critique it still constitutes a complete finding about what representation-level constraints can and cannot buy in this task.

10 Future Work

Three directions fall outside the twelve weeks but follow directly from the design.

Shared/unique factorisation. The triplet term treats publisher information as something to remove. A FactorCL-style decomposition (Liang et al., 2023) would instead split the representation into a component shared across all reports of an event and a component unique to each outlet, keeping both and generating only from the shared part. That is a strictly more informative object than a single scrubbed vector: the unique component is itself a measurement of how a given outlet frames a given story. We treat this as the natural successor to the present objective rather than a competing one.

Full articles rather than excerpts. Our sharpest limitation is that AllSides publishes 85-word excerpts, which caps how much event content there is to retain and therefore how much an event-retention metric can say. Recrawling the source URLs for full text would remove that ceiling and let the QA evaluation carry real weight, at the cost of a crawl and the licensing questions that come with it.

Beyond BART and beyond US media. BART-large is the right backbone for a controlled study on 2,452 examples, but the objective is architecture agnostic and could attach to any encoder. Separately, AllSides' left/centre/right axis is a US-specific, outlet-level construct; whether an event-aware objective transfers to media ecosystems with different partisan structure is an open question our data cannot answer.

11 Broader Impact

Who defines neutral. Our neutral targets are AllSides' human-written roundups, and our stance

labels are outlet-level judgements from the same organisation. A model trained this way does not learn neutrality in the abstract; it learns one editorial organisation's operationalisation of it. We will state this framing explicitly wherever we report results rather than let "neutral" stand unqualified.

Neutrality by deletion is a harm, not just a metric artefact. The failure mode our metric critique identifies is also the ethically serious one. A summariser that scores well by dropping whatever the sources disagree about presents contested reporting as settled and removes exactly the information a reader needs to judge the story. This is why we report the length-normalised rate alongside the raw score, and why we regard a system that lowers framing bias purely by saying less as having failed rather than succeeded.

Dual use of the leakage probe. The probe we use to show that publisher identity has been removed is, run the other way, a stylometric attribution tool for news text. We are building it on published outlets with public bylines, where attribution is not sensitive, but the same method applied to anonymous or pseudonymous writing is a deanonymisation technique. We note this because our contribution is partly the probe protocol itself.

Automation bias. A summary presented as neutral invites more trust than the same text presented as one outlet's account. Any deployment would need to surface its sources rather than replace them.

12 Conclusion

We propose to constrain a multi-source summarisation encoder to organise its latent space around events rather than publishers, using a triplet objective whose negative deliberately shares a publisher with the anchor. The premise is measured rather than assumed: on the NeuS corpus, publisher identity is lexically recoverable at roughly twice its baseline while stance does not transfer across held-out outlets at all. Our audit also found that the field's framing-bias metric is largely a length proxy and that the standard split shares almost all of its test outlets with training, so we contribute a length-normalised metric and a publisher-disjoint protocol alongside the objective. Whether the triplet term buys lower publisher recoverability without costing ROUGE is the question the project answers; a null result, read against the metric critique, is still a complete finding about what representation-level constraints can and cannot do for this task.

13 Timeline

Weeks	Milestone
1–2	Data pipeline, splits, probe protocol *(complete)*
3–4	NeuS baseline reproduced; metric analysis written up
5–6	Triplet objective implemented; λ/margin sweep started
Mid-submission: baseline + first triplet result + metric analysis	
7	Leakage probes on both splits
8–9	Ablations, seed variance, QA check
10–11	Cross-source pretraining extension (compute permitting)
12	Final report and code release

Table 3: Planned schedule. Weeks 1–2 are already complete. The mid-submission checkpoint falls after week 6, by which point the baseline and a first triplet result exist.

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